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Knee joint physical therapy device

Abstract

A device for facilitating physical therapy exercises of the knee having a cylindrical or semicylindrical body having a rigid core, padding, and a vinyl cover. A strap is attached to the body at each end and may have one or more handles for grasping by a patient.

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Background/Summary

FIELD OF THE INVENTION

(1) This disclosure relates generally to physical therapy. More specifically, it relates to a device for facilitating physical therapy exercises for the knees.

BACKGROUND

(2) Physical therapy is often used to treat knee problems, including rehabilitation of the knee after injury or surgery. Physical therapy exercises are often performed using whatever equipment is already available, such as ropes, bands, or towels. Because these devices are not designed specifically for the required exercises, they are often cumbersome or uncomfortable to use. For example, a rolled-up towel may be used to support a joint during exercise, however, the towel may

be too soft to provide adequate support or may come unrolled during the exercise. Patient rehabilitation may be hindered due to the inadequate equipment itself or by frustration caused by the inadequate equipment.
(3) It would therefore be advantageous to have a device which avoids these and other drawbacks of existing devices.
SUMMARY OF THE INVENTION
(4) A knee joint physical therapy device according to embodiments comprises a cylindrical body having a rigid or semi-rigid core, and a cover surrounding the core, and at least one strap attached to the cylindrical body, the at least one strap comprising a first strap end extending from a first end of the cylindrical body and a second strap end extending from a second end of the cylindrical body. The device may also comprise padding surrounding the core and between the core and the cover. The core may comprise a hollow cylinder which may be made from a polymer or may comprise a solid cylinder made of a polymer foam. The first strap end and the second strap end may comprise one or more handles which may be formed by a loop of the strap material. The device may further comprise a retaining band attached to the cylindrical body for passing a foot or both feet of a patient therethrough.
(5) A knee joint physical therapy device according to embodiments comprises a semicylindrical body having a rigid or semi-rigid core, and a cover surrounding the core, and at least one strap attached to the semicylindrical body, the at least one strap comprising a first strap end extending from a first end of the semicylindrical body and a second strap end extending from a second end of the semicylindrical body. The device may also comprise padding surrounding the core and between the core and the cover. The core may comprise a hollow semicylinder which may be made from a polymer or may comprise a solid semicylinder made of a polymer foam. The first strap end and the second strap end may comprise one or more handles which may be formed by a loop of the strap material. The device may further comprise a retaining band attached to the semicylindrical body for passing a foot or both feet of a patient therethrough.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** depicts a perspective view of a knee joint physical therapy device according to one embodiment.
(2) FIG. **2** depicts a cutaway view of the knee joint physical therapy device of FIG. **1**.
(3) FIG. **3** depicts a perspective view of a knee joint physical therapy device according to another embodiment.
(4) FIG. **4** depicts a cutaway view of a knee joint physical therapy device of FIG. **3**.
DETAILED DESCRIPTION
(5) Traditional physical therapy methods make use of existing devices which may not be optimal for performing rehabilitation exercises. As a result, patient rehabilitation may be suboptimal.
(6) To solve these and other problems with existing devices, a knee joint physical therapy device is provided herein.
(7) A knee joint physical therapy device as described herein provides a device which provides rigid, yet comfortable support for certain exercises and provides one or more handles on a pair of straps for facilitating other exercises.
(8) FIG. **1** depicts a perspective view of a knee joint physical therapy device **100** according to one embodiment. Knee joint physical therapy device **100** is comprised of a cylindrical body **102** having a strap **104** attached thereto. Strap **104** is attached to the cylindrical body **102** in a manner that will be described later. Strap **104** has a first strap end **104***a* and a second strap end **104***b*, the first strap end **104***a* attached to a first end **102***a* of the cylindrical body **102** and the second strap end **104***b*

attached to the second end **102***b* of the cylindrical body **102**. First strap end **104***a* and second strap end **104***b* each comprise a plurality of handles **106***a* and **106***b* for grasping by the patient. In other embodiments, physical therapy device **100** may comprise more than one strap. For example, the first strap end may be on a first strap attached at the first end of the cylindrical body and the second strap end may be on the second strap attached at the second end of the cylindrical body.

(9) FIG. **2** depicts a section view of cylindrical body **102**. Cylindrical body **102** is comprised of a cylindrical core **108**, padding **110**, and cover **112**. The primary purpose of cylindrical core **108** is to hold the cylindrical shape of cylindrical body **102**. Cylindrical core **108** may be rigid or semi-rigid and may be comprised of a hollow cylinder or solid cylinder. The core may be extruded, injection molded, or blow molded and may be made out of polyvinylchloride (PVC), polystyrene, polyethylene or any other suitable polymer. The core may be formed of polystyrene, polyethylene or any other polymer foam which may be expanded, extruded, or otherwise formed.

(10) Padding **110** surrounds cylindrical core **108** and may be comprised of any compliant organic or inorganic material such as stuffing, ethylene-vinyl acetate (EVA) foam, polyurethane foam, polyethylene foam, neoprene foam, urethane foam, vinyl foam, silicone foam, PVC foam, memory foam, etc. Padding **110** provides a small amount of cushion for comfort without compressing too much and losing the support of the cylindrical core **108**.

(11) Cover **112** completely surrounds padding **110** and cylindrical core **108** around the circumference and on both ends. Cover **112** is made of a leather, vinyl, or other suitable material to provide a hygienic, easy to clean surface for the device. Cover **112** may be made from one or more pieces attached to one another such as by sewing.

(12) Strap **104** may be attached to cylindrical body **102** by being sewn to cover **112**. Strap **104** may be attached to both ends of cylindrical body **102** as well as along the length of one side of the cylindrical body. Strap **104** may be made of nylon webbing, polyurethane webbing, polyester webbing, or any other suitable material. Strap **104** comprises a plurality of handles **106***a* and **106***b* on either side on the strap. Handles **106***a* are at the ends of strap **104** and handles **106***b* are displaced between handles **106***a* and the cylindrical body **102**. These handles and more may be provided, thus allowing the patient to grasp whichever handles are preferable.

(13) The handles may be made from a loop of the strap material or may be a separate piece attached to the strap. For example, the handles may be made of a plastic or metal loop attached to the strap by sewing, rivets, etc. In the embodiment shown in FIG. **1**, handles **106***a* and **106***b* are formed by folding strap **104** back on itself and securing the two pieces of strap at a first location **114***a* and a second location **114***b*.

(14) A retaining band (not shown in FIG. **1**; See e.g., FIG. **3**) may be included which is comprised of a flexible material attached to the cylindrical body **102** and is designed to help hold cylindrical body **102** on a patient's foot. The retaining band is attached to cylindrical body **102** at two or more places, creating at least one opening for passing a foot or both feet of a patient therethrough. The retaining band may be made of an elastic or inelastic material, such as nylon webbing, rubber, elastic, etc.

(15) FIG. **3** depicts a perspective view of a knee joint physical therapy device **200** according to another embodiment. Knee joint physical therapy device **200** is comprised of a semicylindrical body **202** having a strap **204** attached thereto. Strap **204** is attached to the semicylindrical body **202** in a manner that will be described later. Strap **204** has a first strap end **204***a* and a second strap end **204***b*, the first strap end **204***a* attached to a first end **202***a* of the semicylindrical body **202** and the second strap end **204***b* attached to the second end **202***b* of the semicylindrical body **202**. First strap end **204***a* and second strap end **204***b* each comprise a plurality of handles **206***a* and **206***b* for grasping by the patient. In other embodiments, physical therapy device **200** may comprise more than one strap. For example, the first strap end may be on a first strap attached at the first end of the semicylindrical body and the second strap end may be on the second strap attached at the second end of the semicylindrical body.

(16) FIG. **4** depicts a section view of semicylindrical body **202**. Semicylindrical body **202** is comprised of a semicylindrical core **208**, padding **210**, and cover **212**. The primary purpose of semicylindrical core **208** is to hold the semicylindrical shape of semicylindrical body **202**. Semicylindrical core **108** may be rigid or semi-rigid and may be comprised of a hollow cylinder or solid semicylinder. The core may be extruded, injection molded, or blow molded and may be made out of polyvinylchloride (PVC), polystyrene, polyethylene or any other suitable polymer. The core may be formed of polystyrene, polyethylene or any other polymer foam which may be expanded, extruded, or otherwise formed.
(17) Padding **210** surrounds semicylindrical core **208** and may be comprised of any compliant organic or inorganic material such as stuffing, ethylene-vinyl acetate (EVA) foam, polyurethane foam, polyethylene foam, neoprene foam, urethane foam, vinyl foam, silicone foam, PVC foam, memory foam, etc. Padding **210** provides a small amount of cushion for comfort without compressing too much and losing the support of the semicylindrical core **208**.
(18) Cover **212** completely surrounds padding **210** and semicylindrical core **208** around the circumference, flat side, and on both ends. Cover **212** is made of a leather, vinyl, or other suitable material to provide a hygienic, easy to clean surface for the device. Cover **212** may be made from one or more pieces attached to one another such as by sewing.
(19) Strap **204** may be attached to semicylindrical body **202** by being sewn to cover **212**. Strap **204** may be attached to both ends of semicylindrical body **202** as well as along the length of one side of the semicylindrical body. Strap **204** may be made of nylon webbing, polyurethane webbing, polyester webbing, or any other suitable material. Strap **204** comprises a plurality of handles **206***a* and **206***b* on either side on the strap. Handles **206***a* are at the ends of strap **204** and handles **206***b* are displaced between handles **206***a* and the semicylindrical body **202**. These handles and more may be provided, thus allowing the patient to grasp whichever handles are preferable.
(20) The handles may be made from a loop of the strap material or may be a separate piece attached to the strap. For example, the handles may be made of a plastic or metal loop attached to the strap by sewing, rivets, etc. In the embodiment shown in FIG. **3**, handles **206***a* and **206***b* are formed by folding strap **204** back on itself and securing the two pieces of strap at a first location **214***a* and a second location **214***b*.
(21) Retaining band **216** is comprised of a flexible material attached to the semicylindrical body **202** and is designed to help hold semicylindrical body **202** on a patient's foot. Retaining band **216** is attached to semicylindrical body **202** at two or more places, creating at least one opening for passing a foot or both feet of a patient therethrough. Retaining band **216** may be made of an elastic or inelastic material, such as nylon webbing, rubber, elastic, etc.
(22) In use, the knee joint physical therapy device may be used for multiple physical therapy exercises designed to assist users with knee problems. One such exercise is known as a heel slide. The patient sits on a flat surface and places the cylindrical body **102** or the flat section of the semicylindrical body **202** against the foot and grasps a handle of the strap (**106***a*, **106***b*, **206***a*, **206***b*) on each side of the cylinder. The patient then slides the heel toward and away from the body, pulling on the strap and thus assisting the foot to slide as necessary.
(23) Another exercise that can be performed with a knee joint physical therapy device as described herein is a quad set. In the quad set, the patient lies on a flat surface and the cylindrical body (or semicylindrical body flat side down) is placed under the ankle to raise the leg off the floor. The straps and handles are not usually used and are placed out of the way. The knee is then pushed down toward the floor. The padding **110**/**210** (and possibly also the cover **112**/**212**) provide some cushion for the patient while core **108**/**208** is sufficiently rigid to maintain the required support for the leg. The knee joint physical therapy device thus provides comfortable support for the ankle while the exercise is being performed.
(24) Additionally, a knee joint physical therapy device as disclosed herein may also be used to perform a short arc quad exercise. In the short arc quad, the patient lies on a flat surface and the

cylindrical body **102** (or semicylindrical body **202** with the flat side down) is placed under the knee, causing a bend in the knee when the foot is resting on the flat surface. The foot is then raised by straightening the leg at the knee. The handles and straps are not usually used and are placed out of the way. The padding **110**/**210** (and possibly also the cover **112**/**212**) provide some cushion for the patient while core **108**/**208** is sufficiently rigid to maintain the required support for the leg. Thus, the knee joint physical therapy device provides comfortable support for the knee while the exercise is being performed.

(25) It is contemplated that such device could take many forms without departing from the spirit of the invention. Any combination of features shown in the figures or described in this disclosure may be combined and thus the invention is not limited to the specific embodiments shown.

(26) The discussion herein of the present invention is directed to various embodiments of the invention. The term “invention” is not intended to refer to any particular embodiment or otherwise limit the scope of the disclosure. Although one or more of these embodiments may be preferred, the embodiments disclosed should not be interpreted, or otherwise used, as limiting the scope of the disclosure, including the claims. In addition, one skilled in the art will understand that the following description has broad application, and the discussion of any embodiment is meant only to be exemplary of that embodiment, and not intended to intimate that the scope of the disclosure, including the claims, is limited to that embodiment.

(27) Herein, the terms “including,” “consisting of”, and “comprising” are used in an open-ended fashion, and thus should be interpreted to mean “including, but not limited to.” Also, the term “connect” or “connected” where used if at all is intended to mean either an indirect or direct connection. Thus, if a first component connects to a second component, that connection may be through a direct connection or through an indirect connection via other components and connections.

(28) Certain terms are used throughout the description and claims to refer to particular system components and method steps. As one skilled in the art will appreciate, different companies may refer to a component by different names. This document does not intend to distinguish between components that differ in name but not function.

(29) It is to be understood that the disclosure is not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended claims. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

Claims

1. A knee joint physical therapy device, comprising: a cylindrical body having: a rigid or semi-rigid core, and a cover surrounding the core, and at least one strap attached to the cylindrical body, the at least one strap comprising a first strap end extending from a first end of the cylindrical body and a second strap end extending from a second end of the cylindrical body, wherein the core comprises a solid cylinder, and wherein the first strap end comprises two handles and the second strap end comprises two handles.
2. The knee joint physical therapy device of claim 1, further comprising: padding surrounding the core and between the core and the cover.
3. The knee joint physical therapy device of claim 1, wherein the solid cylinder comprises a polymer foam.
4. The knee joint physical therapy device of claim 1, wherein the handles are formed by a loop of the strap material.
5. The knee joint physical therapy device of claim 1, further comprising a retaining band attached to the cylindrical body for passing a foot or both feet of a patient therethrough.
6. A knee joint physical therapy device, comprising: a semicylindrical body having: a rigid or semi-

rigid core, and a cover surrounding the core, and at least one strap attached to the semicylindrical body, the at least one strap comprising a first strap end extending from a first end of the semicylindrical body and a second strap end extending from a second end of the semicylindrical body, wherein the core comprises a solid semicylinder, and wherein the first strap end comprises two handles and the second strap end comprises two handles.
7. The knee joint physical therapy device of claim 6, further comprising: padding surrounding the core and between the core and the cover.
8. The knee joint physical therapy device of claim 6, wherein the solid semicylinder comprises a polymer foam.
9. The knee joint physical therapy device of claim 6, wherein the handles are formed by a loop of the strap material.
10. The knee joint physical therapy device of claim 6, further comprising a retaining band attached to the semicylindrical body for passing a foot or both feet of a patient therethrough.